

The 65th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

701. EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Vasoactive Intestinal Peptide Signaling in Transplant Recipients Modulates Gut Microbiome Diversity and Regulates Graft-Versus-Host DiseaseYiwen Helen Li¹, Shuhua Wang, MD^{1,2}, Edmund K. Waller, MD³¹Emory University School of Medicine, Atlanta, GA²Emory University School of Medicine and Winship Cancer Institute, Atlanta, GA³Department of Hematology and Medical Oncology, Emory University Winship Cancer Institute, Atlanta, GA

Introduction: Allogeneic bone marrow transplantation (allo-BMT) presents a significant challenge in the form of graft-versus-host disease (GvHD), characterized by donor T cell-mediated attacks on recipient organs, leading to considerable morbidity and mortality. Vasoactive intestinal peptide (VIP), an immunosuppressive neuropeptide, plays a crucial role in shaping T cell responses towards the Th2 phenotype while inhibiting proinflammatory cytokine production. Furthermore, VIP-producing enteric neurons contribute to enteric mucosal barrier homeostasis, maintaining optimal gut pH, and orchestrating the delicate balance between beneficial bacteria colonization and resistance against opportunistic pathogens. The potential impact of VIP production by host cells on GvHD regulation remains unclear. Our hypothesis proposes that the absence of VIP production by transplant recipients may exacerbate GvHD, either through increased T cell allo-reactivity, or via secondary modulation of the gut microbiome resulting from submucosal inflammation. **Methods and Results:** We investigated how endogenous production of VIP by host tissues on GvHD modulation in three murine allo-BMT models, where C57BL/6 wildtype (WT) and VIP-knockout (VIPKO) mice were subjected to 11Gy irradiation on day 0 and subsequently transplanted intravenously with 5×10^6 T cell depleted (TCD) bone marrow (BM) in combination with 0, 1×10^6 , or 3×10^6 T cells from MHC mismatched B10.BR, Balb/c or FvB donors. In all three models, WT recipients had better survival than VIPKO recipients. Specifically, in B10BR- B6, Balb/c - B6 and FvB-B6 model, the addition of 3 million donor T lymphocytes to the graft resulted in significantly more GvHD-related mortality in VIP-KO recipients compared with WT recipients, with 0% vs 82%, 18% vs 96%, and 0% vs 78% survival ($p < 0.01$) among WT vs VIP-KO recipients. To elucidate the role of host cell-derived VIP production in T cell alloreactivity and GvHD exacerbation, we performed cytokine profile analysis using both serum and donor-derived lymphocytes from the spleen of transplant recipients. For serum cytokine analysis, VIP-KO recipients had significantly higher levels of IFN γ ($p < 0.05$), TNF α ($p < 0.05$) and IL-1 β ($p < 0.05$) on day 7 post allo-BMT, and significantly higher levels of IL-6 ($p < 0.01$) and IL-10 ($p < 0.05$) on day 14 post allo-BMT compared with WT recipients. For donor-derived lymphocytes cytokine analysis, we observed significantly increased Th1 and Th17 cytokines in VIP-KO recipient, with significantly higher frequencies of CD4+ IL2 ($p < 0.01$) and CD4+ IFN γ ($p < 0.001$) on day 4, significantly higher levels of CD4+ TNF α ($p < 0.05$) and CD8+ TNF α ($p < 0.001$) on day 19, and significantly increased CD8+ TNF α ($p < 0.05$) on day 25 post allo-BMT compared to WT recipients. Immunofluorescence whole-mount imaging section staining demonstrated detectable VIP expression in enteric neurons innervating the entirety of the intestine in FvB-VIP-GFP transplant recipients. Notably, VIP immunofluorescent signal was much higher in allo-BMT recipient intestines compared to syngeneic transplant recipient intestines, demonstrating immunomodulation of VIP-producing enteric neurons in response to local (GVHD-targeted organ) and systemic inflammation (Fig 1). To investigate the gut microbiome profile, we performed 16S ribosomal RNA sequencing and analyzed stool samples from three distinct groups: 1) naïve WT and VIP KO mice, 2) WT and VIPKO mice after co-housing, and 3) WT and VIPKO mice after co-housing followed by allo-BMT. Uniform Manifold Approximation and Projection high-dimensional cluster analysis revealed that VIP-KO mice exhibited a distinct gut microbiome compared to WT mice, regardless of their co-housing or allo-BMT conditioning (Fig 2A, 2B). The heatmap analysis demonstrated that VIP-KO mice had increased lactobacillus species (Fig 2C) and significantly decreased intestinal pH levels ($p < 0.01$) compared to WT mice. Moreover, VIP-KO mice displayed significantly decreased gut microbiome diversity, indicating an inflammatory effect in the gut that influences changes in the microbiome composition. **Conclusion:** Overall, our findings highlight the crucial role of VIP in regulating immune responses and gut microbiome homeostasis, which could have implications for developing strategies to mitigate GvHD in allo-BMT recipients.

Disclosures No relevant conflicts of interest to declare.

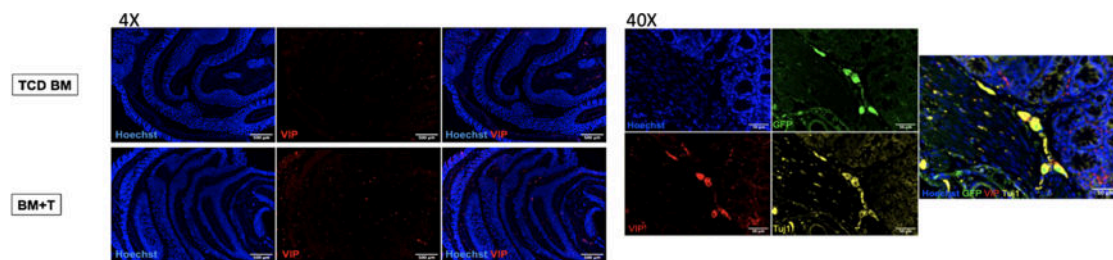


Figure 1. GVHD mice showed increased VIP in enteric neurons. Colon tissue of recipient Vip-EGFP mice (post-transplant Day 10 with FVB TCD BM or BM plus T cells) were stained with anti-GFP (green, Alexa Fluor488), anti-VIP (red, Alexa Fluor 555), anti-Tuj1 (yellow, Alexa Fluor 647), and Hoechst (blue). Mice receiving BM plus T cells (1×10^6) showed increased VIP-secreting neurons detection. Co-staining of GFP, Tuj1, and VIP has confirmed the VIP-secreting enteric neuron staining and VIP antibodies effect. Objective $\times 4$, scale bar 500 μm .

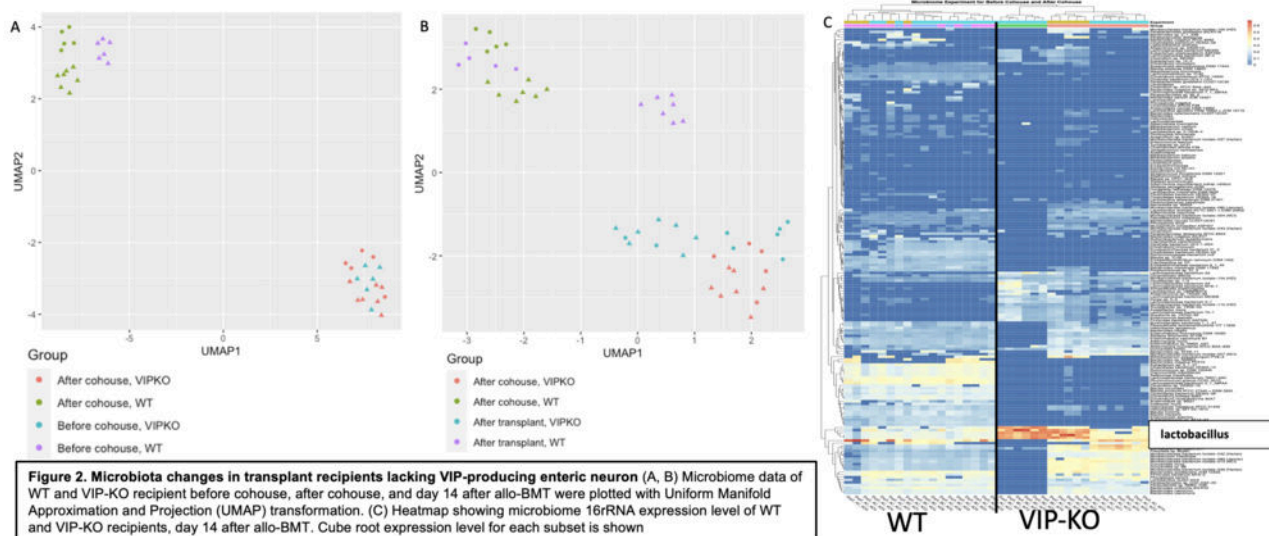


Figure 2. Microbiota changes in transplant recipients lacking VIP-producing enteric neuron (A, B) Microbiome data of WT and VIP-KO recipient before cohousing, after cohousing, and day 14 after allo-BMT were plotted with Uniform Manifold Approximation and Projection (UMAP) transformation. (C) Heatmap showing microbiome 16rRNA expression level of WT and VIP-KO recipients, day 14 after allo-BMT. Cube root expression level for each subset is shown

Figure 1

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